

Simon Gabriel

Charles B. Morrey Jr. Math Fellow
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Research interests

Probability and mathematical statistical physics, with an emphasis on scaling limits of interacting particle systems, stochastic partial differential equations, and critical phenomena.

Academic appointments

Charles B. Morrey Jr. Math Fellow <i>University of California, Berkeley</i> Research in stochastic processes and stochastic partial differential equations; host: Fraydoun Rezakhanlou. Instructor for upper-level analysis courses.	December 2025–present <i>Berkeley, California</i>
Viterbi Postdoctoral Fellow <i>Simons Laufer Mathematical Sciences Institute (SLMath)</i> Member of the programme <i>Recent Trends in Stochastic Partial Differential Equations</i> .	August–December 2025 <i>Berkeley, California</i>
Postdoctoral Researcher <i>University of Münster, Mathematics Cluster of Excellence</i> Research on infinite-dimensional stochastic models in Hendrik Weber’s stochastic analysis group.	October 2023–July 2025 <i>Münster, Germany</i>

Education

PhD in Mathematics <i>University of Warwick</i> Thesis: <i>Critical scaling limits and singular SPDEs</i> . Supervisors: Nikos Zygouras and Paul Chleboun.	2019–2023 <i>Coventry, United Kingdom</i>
MSc in Mathematics <i>Technical University of Munich</i>	2018–2019 <i>Munich, Germany</i>
BSc in Mathematics, Minor in Computer Science <i>Technical University of Munich</i>	2014–2018 <i>Munich, Germany</i>
Visiting Student in Mathematics <i>National University of Singapore</i>	2017 <i>Singapore</i>

Additional training

Technical AI Safety (six-week course) <i>BlueDot Impact</i>	September–October 2026 <i>Online</i>
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Publications and preprints

Preprints

- [9] **S. Gabriel**, *Equilibrium fluctuations of the weakly asymmetric inclusion process*, 2026. [arXiv:2608.29957](https://arxiv.org/abs/2608.29957).
- [8] **S. Gabriel** and M. Tempelmayr, *Weak universality for stochastic reaction-diffusion models with long-range correlated noise*, 2026. [arXiv:2605.28677](https://arxiv.org/abs/2605.28677).
- [7] L. Ferdinand and **S. Gabriel**, *Discrete RG flow of a hierarchical singular SPDE*, 2026. [arXiv:2603.17177](https://arxiv.org/abs/2603.17177).
- [6] **S. Gabriel** and T. Rosati, *Fluctuations in the weakly coupled 4D Anderson Hamiltonian*, 2026. [arXiv:2602.22509](https://arxiv.org/abs/2602.22509).
- [5] **S. Gabriel**, *Modulated Poisson–Dirichlet diffusions arising from inclusion processes with a slow phase*, 2025. [arXiv:2507.13799](https://arxiv.org/abs/2507.13799).

Peer-reviewed publications

- [4] **S. Gabriel**, T. Rosati, and N. Zygouras, *The Allen–Cahn equation with weakly critical random initial datum*, *Probability Theory and Related Fields* 192, 1373–1446 (2025). [doi](#).
- [3] P. Chleboun, **S. Gabriel**, and S. Grosskinsky, *Size-biased diffusion limits and the inclusion process*, *Electronic Journal of Probability* 29, paper 65 (2024). [doi](#).
- [2] **S. Gabriel**, *Central limit theorems for the $(2+1)$ -dimensional directed polymer in the weak disorder limit*, *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques* 59(4), 1984–2016 (2023). [doi](#).

- [1] P. Chleboun, **S. Gabriel**, and S. Grosskinsky, *Poisson–Dirichlet asymptotics in condensing particle systems*, *Electronic Journal of Probability* 27, paper 155 (2022). doi.

Awards

Faculty Thesis Prize	2024
<i>Faculty of Science, Engineering and Medicine, University of Warwick</i>	<i>Coventry, United Kingdom</i>
EPSRC Doctoral Training Partnership Award	2019
<i>University of Warwick</i>	<i>Coventry, United Kingdom</i>

Selected talks and lectures

2026	University of Augsburg — Probability Seminar (invited) National University of Singapore — Statistical Mechanics and Singular SPDEs (invited) UC Berkeley — Student Probability/PDE Seminar (lecture series) Purdue University — Probability Seminar (invited)
2025	UC Berkeley — Probability Seminar (invited) SLMath — SPDE Seminar
2024	Université Lyon 1 and ENS Lyon — Probability Seminar (invited) University of Warwick — Junior Analysis Seminar (invited)
2023	Beijing Institute of Technology — Webinar on Stochastic Analysis (invited) Technical University of Vienna — Probability Seminar (invited) University of Oxford — Workshop on Interacting Particles, Fluctuating Systems, and SPDEs (invited) National University of Singapore — Probability Seminar (invited) Max Planck Institute for Mathematics in the Sciences — Analysis–Probability Seminar (invited)

Research visits

Research visit hosted by Rongfeng Sun	April 2023
<i>National University of Singapore</i>	<i>Singapore</i>

Teaching

Instructor	2026
<i>University of California, Berkeley</i>	<i>Berkeley, California</i>
<i>Introduction to Complex Analysis (Math 185), two sections; Introduction to Analysis (Math 104).</i>	
Teaching Assistant	2020–2021
<i>University of Warwick</i>	<i>Coventry, United Kingdom</i>
<i>Stochastic Analysis (MA482); Mathematical Analysis III (MA258); Probability Theory (MA318).</i>	
Teaching Assistant and Marker	2018–2019
<i>Technical University of Munich</i>	<i>Munich, Germany</i>
<i>Probability Theory (MA2409); Introduction to Probability Theory (MA1401).</i>	

Service and organisation

Co-organiser, GEN–Y Research Workshop in Stochastic Analysis	2024
<i>University of Münster</i>	<i>Münster, Germany</i>
Co-organised with Markus Tempelmayr.	
Co-organiser, Postgraduate Seminar in Mathematics	2020–2021
<i>University of Warwick</i>	<i>Coventry, United Kingdom</i>
Co-organised with Arjun Sobnack.	
Referee	
<i>Annales de l'Institut Henri Poincaré, Probabilités et Statistiques; Communications in Mathematical Physics; Electronic Communications in Probability; Journal of Statistical Physics; and Nonlinearity.</i>	